Queensland University of Technology
Faculty of Science
School of Mathematical Sciences

Computational Modelling of Multilateral Treaties

Stage 2

Candidate: Oriolson Rodriguez Ramirez
Student Number: n12608521
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Supervisory Team

Role	Name	Affiliation
Principal Supervisor	Prof. Michael Bode	School of Mathematical Sciences, QUT
Associate Supervisor	Prof. Jonathan Rhodes	School of Mathematical Sciences, QUT

Principal Supervisor — Prof. Michael Bode Prof. Bode is Deputy Director of Capability Development at SAEF (Securing Antarctica's Environmental Future). His work centres on applied mathematics and conservation, with a focus on decision-making under uncertainty and mathematical biology. His previous work on Antarctic Treaty System (ATS) decision-making and conservation makes him an expert on consensus-based systems.

Associate Supervisor — Prof. Jonathan Rhodes Prof. Rhodes is Director of the Centre for Environment and Society. He specialises in spatial ecology, environmental governance, and the modelling of coupled human–natural systems, bridging quantitative models with real-world policy. His expertise gives him deep insight into the policy implications of environmental decisions, providing a grounding for the treaty-system applications explored in this research.

Chapter 1

Background and Literature Review

1.1 Introduction

Many international insitutions use consensus or unanimity as the voting system choice. This can implicitly caps what this institutions can actually achive. A consensus-based system earns broad legitimacy by granting equal veto power to every member. However, this gain in legitimacy trades off with slow adaptation, ultimately risking institutional inertia and decline. (Gray 2018). The Antarctic Treaty System (ATS) is one of those consensus-driven regimes, negotiated between a set of parties called consultative parties Secretariat of the Antarctic Treaty (2023). This group of countries, together with a second group lacking decision-making power (non-consultative parties), meets yearly at the Antarctic Treaty Consultative Meeting (ATCM). Before each meeting, culsultative parties may submit Working Papers (WP) and Information Papers (IP) up to 45 days before the ATCM starts Secretariat of the Antarctic Treaty (2023). During ATCMs the parties discuss Working Papers and produce outcomes only by consensus. These outcomes take the form of Measures, Decisions, or Resolutions Antarctic Treaty Consultative Meeting (1995). As international relations strain and Antarctic environmental pressures intensify, the risk of losing the capacity to act increases. In 2022, for the first time, the ATCM parties failed to reach consensus on the final report; even as activity has increased, the regime has become less responsive to challenges Gardiner et al. (2025).

In a system based on consensus is or utmost importance to uderstand why it succeds or fails, And which parties drive it or block it, respectively. Existing literature has described the ATS by rates and diversity of outputs, and stucturally by describing who co-aouthors with whom Larissa Lubiana Botelho et al. (2026) and Kumar et al. (2026). But there is no study where the a reconstruction of a directional party intention was reconstructed from the textual records on each issue over time. Even further, that type of measure has not been used to model the strategic behaviour that produces or breaks consensus. This thesis objective is to measure that and to build that model and asks if they contain the strategic intentions made by the consultative parties. It then asks whether these can model historical consensus dynamics.

The approach is a computational pipeline. First, it uses embedding techniques to discover topics, and NLI (Natural Language Inference) techniques for stance extraction. These would allow the construction of the preference measure. Second, it applies NLI to extract the opposition or aversion. It then testes the approach by comparing inputs (i.e. working papers) against observes outcomes. Finally, a Multi-Agent reinforcement learning is trained based on the previous outcomes to simulate the negotiation itself.

1.2 The Antarctic Treaty System as a consensus regime

In the ATS, Consultative Parties (CPs) reach outcomes exclusively by consensus. These discussions take place at the annual ATCM, where outcomes can take any of the following forms, per Secretariat of the Antarctic Treaty (2023, Rule 24):

- **Measures** are adopted texts containing provisions intended to become legally binding; these require approval by the relevant parties' legislative bodies.
- **Decisions** are adopted texts concerning internal organisational matters of the ATCM.
- **Resolutions** are non-binding recommendations or expressions of intent.

There are four types of input:

- **Working Papers** (Secretariat of the Antarctic Treaty (2023, Rule 48)): submitted by CPs and Observers, these require discussion at the ATCM and are the main source of outcomes. They are the only input from CPs that can produce an outcome; as such, they are the most likely place for each country to state its intentions.
- **Secretariat Papers** (Secretariat of the Antarctic Treaty (2023, Rule 49)): prepared and presented by the Secretariat, these produce organisational decisions as output, e.g. the budget.
- **Information Papers** (Secretariat of the Antarctic Treaty (2023, Rule 50)): supporting information for Working Papers or other discussions. They may be presented by CPs, Observers, Non-Consultative Parties (NCPs), and experts.

- **Background Papers** (Secretariat of the Antarctic Treaty (2023, Rule 51)): submitted by ATCM participants but not introduced during the meeting; they exist solely for the record.

Of all four, Working Papers are the only input uniquely submitted by CPs, requiring discussion and not confined to organisational matters. They are therefore the most likely place for countries to express their strategic intentions within this closed decision-making environment.

The consensus system makes the ATS a notable example of international cooperation; however, the same mechanism has become a major challenge as the number of CPs grows. Gardiner et al. (2025) show that the decision-making responsiveness of the ATS is declining; the unprecedented 2022 consensus failure is a clear example.

A notable case of contested consensus occurred in the late 1970s and 1980s, during the negotiations over mineral resources in Antarctica. The Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) was produced and then abandoned in favour of the Madrid Protocol (1991), which prohibited all mineral resource extraction in Antarctica (Jackson 2021). This well-documented episode provides a valuable canary test: a known case with documented positions against which any quantitative stance measure can be validated.

1.3 Quantitative and computational study of International institutions and treaties

In general the quantitative study of international institutions exist long before it was applied to Antarctica. Substantial literature maps the design of treaties and organisations into structured variables for example: their decision rules, dispute-settlement provisions, and enforcement mechanisms. Furthermore, these studies track these variable to to institutional performance and survival (Gray 2018).A more recent trend treats the *text* of international agreements and debates as data. They recover state positions from voting records and speeches (Baturo et al. 2017; Slapin and Proksch 2008). Both approaches show that preferences and design features of treaty regimes can be measured, not mereley described.

The ATS has been analysed quantitatively in two ways: Gardiner et al. (2025) does performance accounts that measure the regime in aggregate, counting inputs and outputs, and the time legally binding measures take to enter into force. It concludes that responsiveness to challenges in the Antarctic region is declining. The second methodology is structural (Larissa Lubiana Botelho et al. (2026) and Kumar et al.

(2026)): mapping who collaborates with whom through co-authorship networks of ATCM documents. Both analyses are very valuable, and in this project both results are going to be used as an external check: the network structure as a replication target for the authorship layer, and the outcome series as a prediction target in Chapter 2. Neither, however, extracts topics from WPs in an unsupervised manner, nor yields a directional, country- and topic-level, time-resolved measure of policy preference.

1.4 Measuring policy preference from political text

Positional measures extracted from political texts are a well-developed field with two paradigms. The first paradigm is scaling and ideal-point methods, which recover a latent positional dimension without a pre-specified axis, i.e. there is no need for a regulate vs permissive comparison. These methods tipically rely on word counts (i.e. frequencies) as their main underlying principle. This is the case of "Wordfish", introduced by Slapin and Proksch (2008) as an unsupervised positional scaling of party text. Lauderdale and Herzog (2016) addressed the problem that legislators talk about different things in different debates; they made a per-topic, per-debate scaling. On the other hand, Vafa et al. (2020) produced a probabilistic framework to place ideal points and the language of different actors. The closest dataset to the ATCM corpus was the one used by Baturo et al. (2017) to derive a state-level preference using United Nations (UN) debate speeches. Finally, a recent review by Parschan and Jakob (2025) condenses all these approaches in one paper. The second paradigm uses embeddings as its main tool. Rheault and Cochrane (2019) show that word-embedding geometry recovers ideological placement using parliamentary data. On the other hand, Rodriguez and Spirling (2022) provide a comprehensive analysis of what works and what does not when analysing political text using embedding techniques. They find embeddings a useful technique for those purposes but caution that the choice of embeddings strongly affects applied inference, and that the results must be validated. Finally, Licht (2023) and Chen et al. (2024) extend this approach to cross-lingual and multilingual settings.

Here, we propose to use both paradigms in the same pipeline but at different stages. This thesis differs from the ideal-point methods in an important way: ideal-point models recover a single stable latent position per actor, whereas the presented project requires a topic-conditioned stance on a given axis. The proposal in this thesis allows the parties to hold different positions on different topics through time. The topic structure is discovered using the embedding-based approach of Grootendorst (2022), which couples transformer representations with density clustering and class-based TF-IDF. The stance measure is then estimated with reference to the models of Burnham (2023) and Abercrombie and Batista-Navarro (2022). This thesis proposes to use discriminative transformer-NLI models, which take a **premise** and a **hypothesis**. These NLI models output one of the prescribed classes and $P(\text{entailment} \mid \text{hypothesis})$, then the class with the highest probability is selected. The final results can be validated directly using the CRAMRA canary test (Jackson 2021), or indirectly by checking which countries cluster around $(\text{topic}, \text{stance})$ pairs and comparing this to the Larissa Lubiana Botelho et al. (2026) network results.

1.5 Modelling consensus, coalition formation and negotiation

There two traditional group of models how groups reach concensus. The first is the mosts ellegant but are the synchronization and opinion-dynamics models, like Kuramoto and opinion-changing-rate proposed by Leonardo L. Botelho (2022) which represent agreement as an coupled-oscillator convergence. These models does not contain strategic preference, no veto and not learning, therefore, it cannot simulate the bargaining behaviour that occurs in concensus based systems. The second group of models are strategic and multi-agent. In the case of coallition formatiion there is substantial literature (Sarkar et al. 2022), and negotiation simulations has been recently reated as multi-agent reinforcement learning. For example, Chang (2021) used deep reinforcement learning to model multi-issue bargaining, Bachrach et al. (2020) model team formation using the same approach,

1.6 Summary of Literature Gap

The literatures reviewed above have matured along a common arc: qualitative and historical accounts of consensus institutions, then descriptive measurement of their outputs (Gardiner et al. 2025), then structural mapping of who collaborates with whom (Larissa Lubiana Botelho et al. 2026; Kumar et al. 2026), and most recently formal and learning-based models of negotiation. Each step left a layer unaddressed. Descriptive accounts measure *what* the regime produces but not *who* wanted what; structural accounts are undirected and unsigned, so they record collaboration without content or direction; the text-as-data tradition recovers positions, but as stable latent ideal points rather than topic-conditioned, time-resolved stances (Slapin and Proksch 2008; Lauderdale and Herzog 2016); and formal models of consensus either treat agreement as passive convergence, with no veto and no status quo (DeGroot 1974; Hegselmann and Krause 2002), or, where they are strategic (Tsebelis 2002) and learning-based (Moghimifar et al. 2024; Bolleddu 2025), run on synthetic or stated-platform preferences rather than an institution's own deliberative record.

The gap is therefore the missing coupling between these layers: no existing work extracts a validated, directional, signed measure of national policy preference — country by topic and over time — from a consensus regime's own textual record, and uses it to drive a strategic model in which parties act as the veto players the decision rule makes them, validated against the regime's realised outcomes. This thesis supplies that coupling for the Antarctic Treaty System: Chapter 1 constructs the support layer, Chapter 2 adds the opposition layer and tests the combined structure against observed outcomes (Gardiner et al. 2025), and Chapter 3 embeds both in a strategic simulation capable of institutional counterfactuals.

Chapter 2

Program Design

2.1 Research Questions

Research Questions

RQ1. How do **affinity dynamics** — patterns of positive alignment and coalition-building — shape the emergence and stability of consensus within the Antarctic Treaty System?

RQ2. How do **aversion dynamics** — veto behaviour, strategic defection, and antagonistic positioning — threaten or reconfigure treaty equilibria?

RQ3. What computational simulation framework can reproduce empirically observed ATS decision outcomes, and what generalisable principles does it reveal for multilateral treaty design?

2.2 Theoretical Framework

The project is grounded in three complementary theoretical traditions:

1. **Non-cooperative game theory** — models individual state incentives and deviation conditions (Nash 1951; Osborne and Rubinstein 1994).
2. **Opinion dynamics / consensus models** — captures iterative belief convergence among parties (DeGroot 1974).
3. **Agent-based simulation** — provides a computational substrate for running counterfactual treaty scenarios (Epstein and Axtell 1996).

2.3 Research Design Overview

The project is structured into three interconnected papers, each targeting one research question and one modelling contribution.

Table 2.1: Paper structure and milestones

Paper	Focus	Method	Target venue
1	Affinity dynamics	Network analysis, coalitional GT	TBD
2	Aversion dynamics	Veto-player model, ABM	TBD
3	Simulation framework	Agent-based model, RL	TBD

2.4 Paper 1 — Affinity Dynamics in Multilateral Treaties

2.5 Paper 2 — Aversion Dynamics and Consensus Breakdown

2.6 Paper 3 — Integrated Simulation Framework

2.7 Data Sources and Ethics

The primary data sources for this project are:

- Antarctic Treaty Consultative Meeting (ATCM) final reports and working papers (publicly available at https://www.ats.aq/).
- State-level foreign policy position datasets (e.g., UN voting records).
- Published network and voting data from the international relations literature.

All data are publicly available institutional records; no human-subjects research is involved and no ethics clearance is required beyond standard QUT research integrity protocols.

2.8 Timeline

Table 2.2: Indicative project timeline

Phase	Activity	Target Date
1	Literature review & theoretical framework	Month 6
2	Paper 1: data collection & analysis	Month 12
3	Paper 1: writing & submission	Month 15
4	Paper 2: model development & calibration	Month 18
5	Paper 2: writing & submission	Month 21
6	Paper 3: simulation framework development	Month 27
7	Paper 3: writing & submission	Month 30
8	Thesis integration & write-up	Month 36
9	Thesis submission	Month 38

2.9 Expected Contributions

1. A formal computational model of consensus dynamics in the ATS, the first of its kind at this level of institutional specificity.

2. A transferable agent-based simulation framework applicable to any consensus-based multilateral treaty.

3. New theoretical constructs (affinity/aversion indices) operationalising previously qualitative political science concepts.

4. Policy-relevant insights into treaty design features that improve stability under strategic pressure.

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Appendix A

Supplementary Material

A.1 Notation and Conventions

Table A.1: Notation used throughout the document

Symbol	**Meaning**
N	Set of treaty parties (agents), $\|N\| = n$
$i, j \in N$	Individual state actors
$s_i \in S$	Strategy of party i
u_i	Utility function of party i
α_{ij}	Affinity score between parties i and j
β_{ij}	Aversion score between parties i and j
$\mathbf{W}$	DeGroot influence/weight matrix
$x_i(t)$	Position/opinion of party i at time t

A.2 Acronyms

ATS Antarctic Treaty System

ATCM Antarctic Treaty Consultative Meeting

ATP Antarctic Treaty Party

ABM Agent-Based Model

RL Reinforcement Learning

GT Game Theory

IR International Relations

QUT Queensland University of Technology

A.3 Candidate's Publication Record

No publications to date; in preparation.